EV Charging Network Analytics



Power BI dashboard on revenue, demand and reliability across 10 Indian cities

Individual Project | Power BI | October 2026

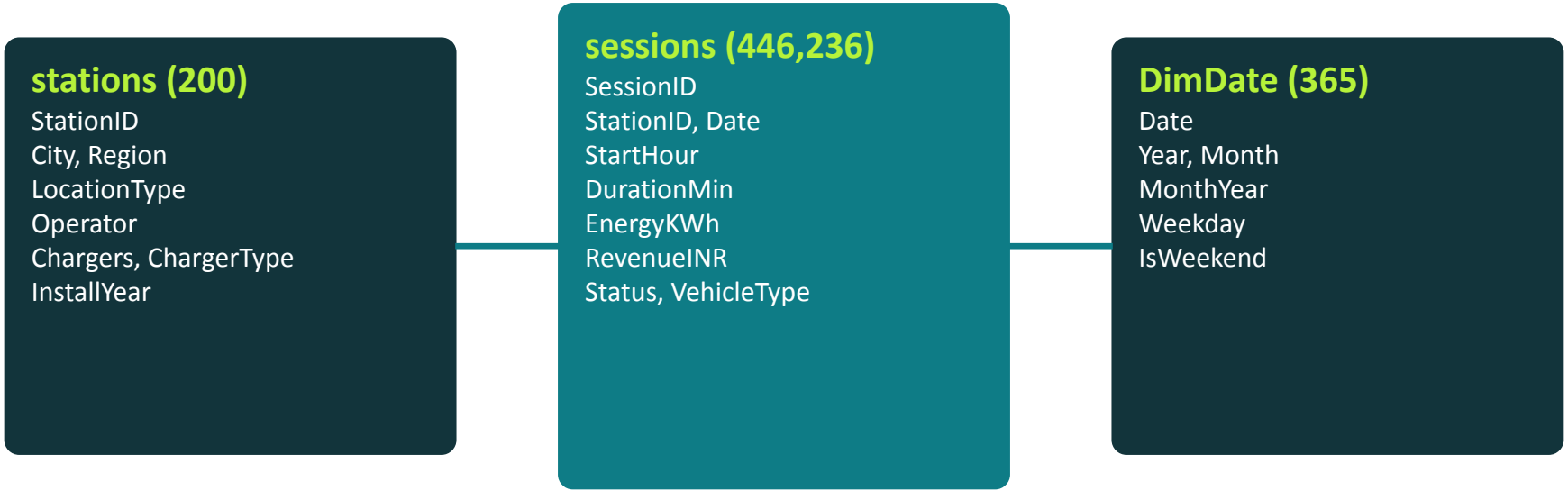
Why this analysis matters

1. **Problem** Operators cannot see where revenue comes from or where service fails.

2. **Scope** 200 stations, 10 cities, 5 operators, Oct 2025 to Sep 2026.

3. **Goal** Find growth levers, peak-demand patterns and reliability gaps.

Dataset and data model



Star schema with one-to-many relationships on StationID and Date

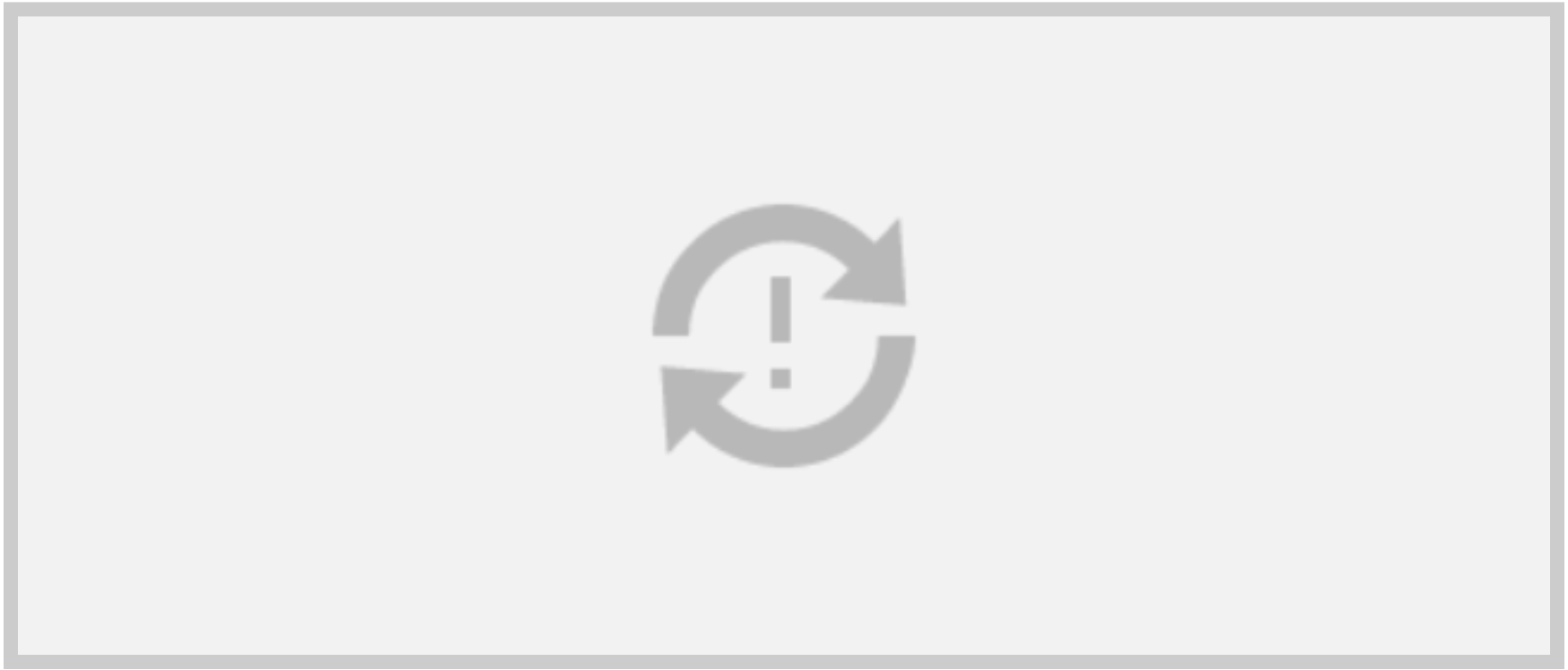
Synthetic data generated with gen_data.py (seeded). Not real operator data.

Network at a glance

₹20.4 Cr	9.0M	446K	4.0%
Total revenue	kWh delivered	Charging sessions	Failure rate

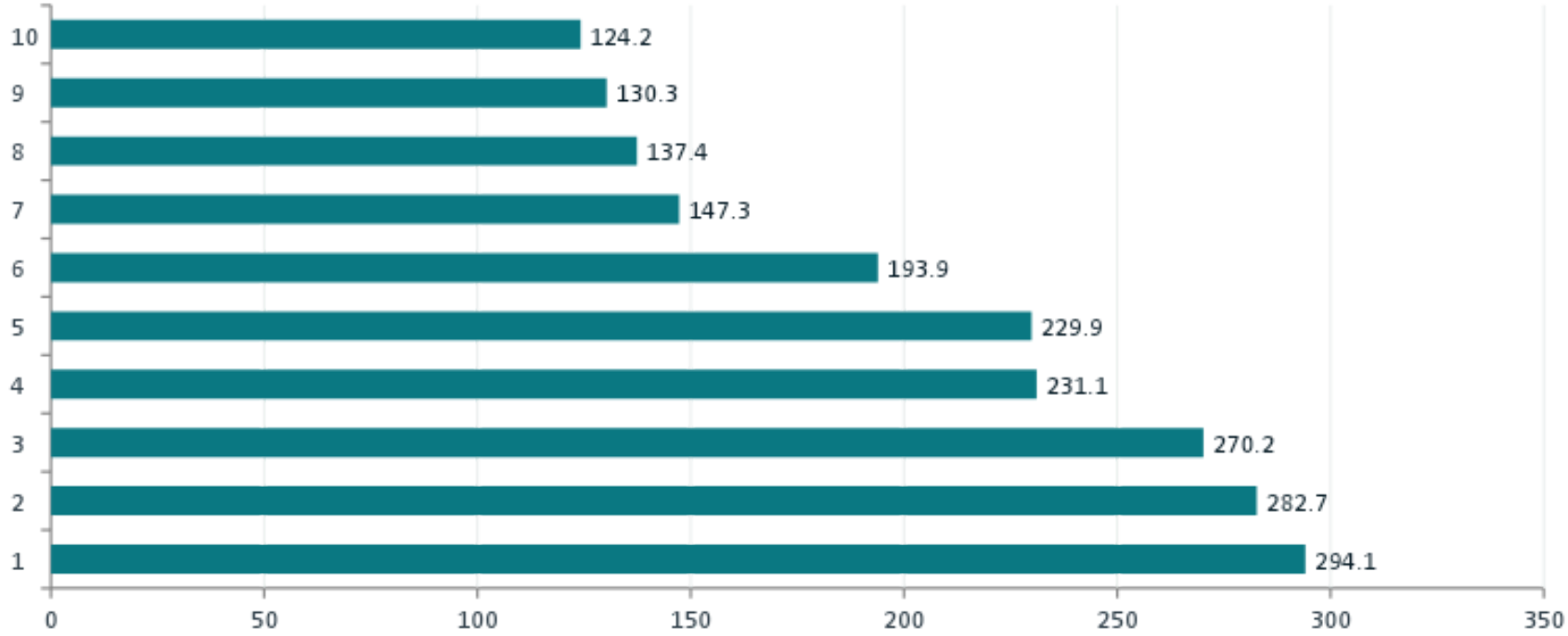
Revenue counts completed sessions only. Average of about ₹470 per completed session.

Revenue grew about 36% in 12 months



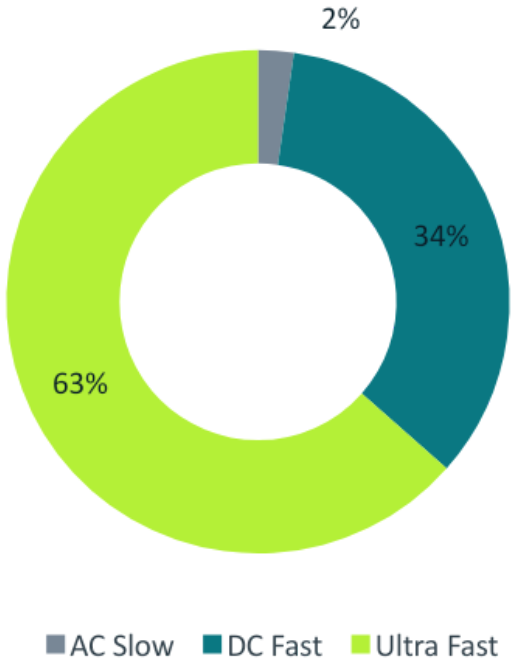
Monthly revenue in ₹ lakh. Oct 2025 ₹144 lakh to Sep 2026 ₹195 lakh.

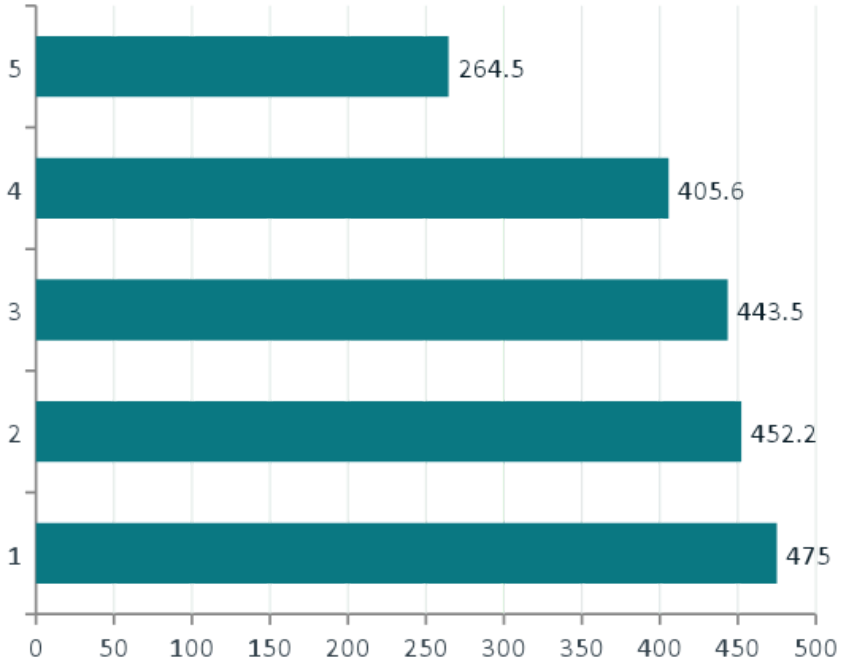
Ahmedabad, Pune and Bengaluru lead on revenue



Revenue by city in ₹ lakh. Chennai, Jaipur and Delhi trail despite large markets.

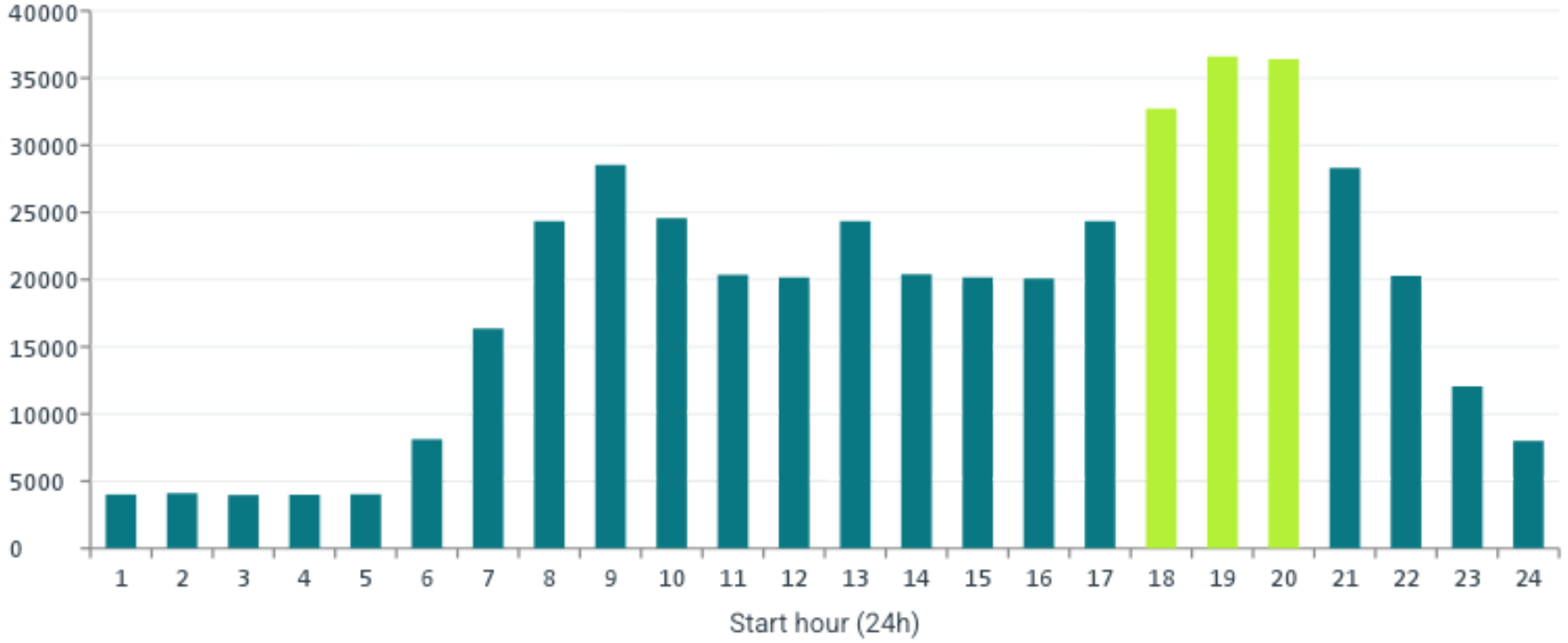
Ultra Fast chargers earn 63% of revenue





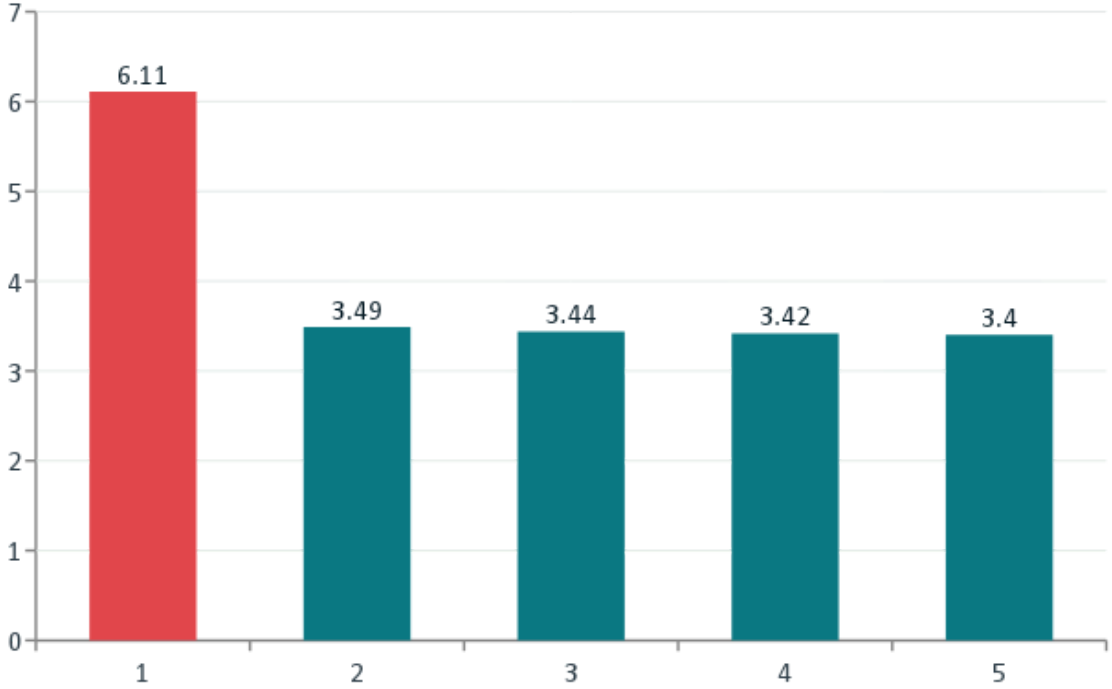
Left: revenue share by charger type. Right: revenue by location type in ₹ lakh.

Demand peaks between 5 and 8 PM



Sessions by start hour. A smaller morning peak appears at 8 AM.

ChargeZone fails almost twice as often



6.1%

ChargeZone failure rate vs about 3.4% for every other operator

Failure rate = failed sessions / total sessions, by operator.

Recommendations

Expand Ultra Fast

Add Ultra Fast chargers on highways and in top-revenue cities.

Fix reliability

Review ChargeZone uptime and maintenance contracts.

Shift peak load

Use off-peak pricing to ease the 5 to 8 PM rush.

Rebalance sites

Favour workplace and parking sites over residential AC chargers.

Thank you

Tools: Power BI Desktop, Power Query, DAX, Python
Code, data and report on GitHub